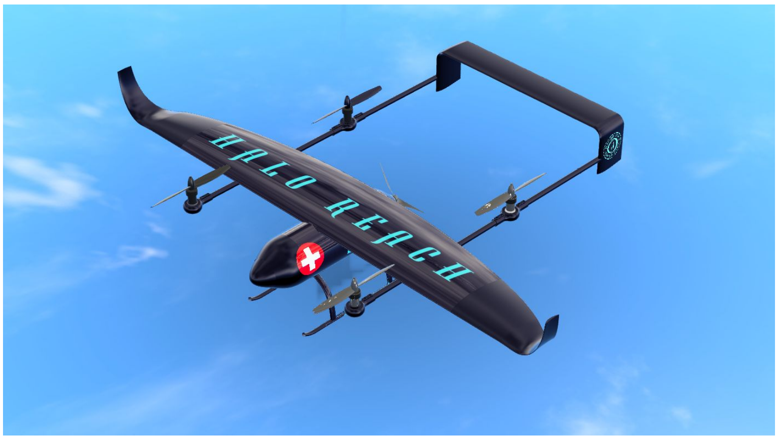
HALO REACH: Empennage and Lateral Control Design

UAS for Disaster Relief - AE3 2024



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Abstract

This report details the empennage configuration, static stability, and lateral dynamic steadiness of HALO's REACH UAV. An inverted U-tail was selected, and a grid search algorithm was designed to determine the optimal size for the vertical and horizontal stabilisers, achieving minimal mass with volume coefficients of 0.69 for the horizontal tailplane and 0.04 for vertical tailplane. The UAV's ability to dampen yaw, roll, and pitch disturbances was verified, ensuring static stability. Additionally, the lateral dynamic behaviors: roll subsidence, dutch roll, and spiral were examined to confirm dynamic stability and control at various fuel loads and cruising velocities. In all tested scenarios, the REACH UAV demonstrated both static and lateral dynamic stability with its mass optimised empennage.

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Nomenclature

α	Angle of attack
β	Sideslip angle
η_h	Dynamic pressure ratio
$\frac{\partial\epsilon}{\partial\alpha}$	Downwash derivative
ρ_B	Density of the boom
ρ_w	Density of the wing
ζ	Damping ratio
A	Wing aspect ratio
b	Wing span
B_A	Boom cross sectional area
B_h	Maximum height of the boom cross section
B_l	Boom length
B_y	Boom centre line span wise location
B_{fl}	Boom length forward of wing leading edge or trailing edge
B_{xm}	Longitudinal distance of leading edge forward of the c.g. for the booms
B_{zm}	Boom vertical distance down from the wing
B_{zm}	Vertical distance of boom centre-line below the c.g
$C_{L_{\alpha,h}}$	Lift curve slope of the horizontal stabiliser
$C_{L_{\alpha,w}}$	Lift curve slope of the wing
C_{L_α}	Lift curve slope of the aircraft
C_{l_β}	Coefficient moment around the x-axis with respect to sideslip
$C_{m,w,T,f,h}$	Coefficient of moments produced by the wing, engine thrust, fuselage and booms when the aircraft is in steady level flight at cruise.
C_{m_α}	Coefficient of pitching moment with respect to angle of attack
$C_{m_{\alpha,w,f,B,h}}$	Coefficient of pitching moment with respect to angle of attack from the wing, fuselage, booms and horizontal stabiliser respectively
C_m	Coefficient of pitching moment
$C_{n_{\beta,f}}$	Fuselages contribution to the yawing moment derivative with respect to sideslip
C_{n_β}	Coefficient moment around the z-axis with respect to sideslip
$C_{Y_{\beta,v}}$	Coefficient of lateral force with respect to sideslip for the vertical stabiliser
C_{Y_β}	Side force with respect to sideslip
d_f	Diameter of the fuselage
K	Boom fineness ratio
l_f	Length of the fuselage
l_h	Distance between the wing and horizontal stabilisers aerodynamic centres

l_v	Moment around the x-axis with respect to sideslip velocity
M_e	Mass of the empennage
n_v	Moment around the z-axis with respect to sideslip velocity
S_h	Horizontal tailplane planform area
S_v	Vertical tailplane planform area
S_w	Wing planform area
t_x	X position of the tailplanes trailing edge from the fuselage nose
V	Cruise velocity
v	Sideslip velocity
w	Width of the booms or fuselage when used in the Multhopp method
w_n	Natural frequency
w_x	X position of the wings leading edge from the fuselage nose
x_{ac}	Longitudinal position the aerodynamic centre
x_{cg}	Longitudinal position of the centre of gravity from the fuselage nose
x_{np}	Longitudinal positon of the aircraft neutral point
Y_v	Side force with respect to sideslip velocity
GTOW	Gross Takeoff Weight
HALO	Humanitarian Aid Logistics Operations
REACH	Remote Emergency Assistance and Communications Hub
UAV	Unmanned Aerial Vehicle
VTOL	Vertical Take-Off and Landing

1 Context and Motivation

With the onset of a catastrophic disaster, transport links such as roads, train tracks, and runways are often damaged. This hampers emergency services as they struggle to reach those affected. This was especially evident on February 6, 2023, when a series of high-magnitude earthquakes struck Turkey and Syria. These earthquakes resulted in the confirmed deaths of 50,783 people in Turkey and 8,476 people in Syria. The Turkish government cited disruptions to communications and damaged roads as significant challenges in managing this tragedy [1].

In response to such challenges, the Vertical Flight Society has established criteria for a modular, multi-mission Vertical Take-off and Landing (VTOL) Unmanned Aerial Vehicle (UAV) to assist emergency services in delivering life saving relief [1]. Based on the provided criteria, the UAV must be capable of performing the following tasks:

1. Take-off and land vertically from the deck of a ship in high winds and gusty conditions.
2. Cruise to and from the site of a disaster.
3. Act as a long endurance communications relay during an extended loiter flight phase.
4. Land vertically within a disaster stricken region to deliver relief supplies.
5. Features a modular add-on system allowing two people to swap payloads for either long-endurance missions or supply delivery missions in less than ten minutes.

To fulfill the brief, HALO (Humanitarian Air Logistics Operations) has designed a lift-plus-cruise VTOL UAV that can undertake both long endurance and supply missions called REACH (Remote Emergency Assistance and Communications Hub). Detailed flight profiles for these missions are depicted in Figure 9 and Figure 10 in Appendix A. With key specifications related to the UAV's target parameters outlined in Table 7 in Appendix A.

2 Report Outline

Since the aircraft is unmanned, it is expected to operate autonomously for most of its mission profile. Therefore, ensuring the UAV's stability and controllability is crucial, with the empennage being a key contributing factor. Thus, this report details the design of the empennage and examines the static and dynamic stability of the UAV's open-loop response to ensure the aircraft is stable and adequately controllable for autonomous missions.

3 Empennage Design

From the conceptual design [2], the UAV was chosen to be a twin boom propeller pusher aircraft with a rear mounted empennage. As a result, the empennage should be responsible for creating forces around the lateral axis and vertical axis to ensure trim, stability and control.

A list of potential tailplane arrangements, based on a review of past aircraft with similar designs, is shown in Figure 3.

The use of a conventional boom-mounted tailplane would result in good deep stall characteristics due to the low position of the horizontal tailplane relative to the wing [3]. However, the presence of a pusher propeller would cause the wake generated to interfere with the horizontal tailplane, exposing it to higher dynamic pressure than the free stream [4], along with any associated turbulence from the propeller, which would reduce control authority. Consequently, this configuration was not considered further.

Both the inverted U and inverted V arrangements could position the tailplane outside the propeller wake but would also make it susceptible to deep stall. The inverted V-tail design has advantages such

Figure 3. Possible empennage configurations for a boom mounted tailplane: Conventional (left), inverted U (middle) and inverted V (right).

as structural rigidity, reduced interference drag, and simpler maintenance and production due to fewer actuators and surfaces [5]. However, it also has drawbacks, including coupling between pitch and yaw control inputs and potentially greater mass [5]. For the REACH UAV, the booms are spaced widely in the span-wise direction to accommodate the large vertical rotors and pusher propeller, which may required an oversized V-tail to avoid the propeller wake.

The inverted U-tail was chosen. Although it is not as structurally rigid as the V-tail and prone to higher drag, it allows for the decoupling of pitch and yaw control, simplifying the controller design, further discussed in [6]. This was deemed important as the drone is expected to fly autonomously, and with the GTOW being very restrictive, higher priority was given to components that could offer mass savings. To mitigate deep stall tendencies, the vertical tailplane position will be adjustable, allowing it to extend below the booms, as shown in Figure 6 and discussed in section 3.2.1.

3.1 Horizontal Tailplane Design

The horizontal tailplane ensures longitudinal stability and trim by balancing the moments from other aircraft components. Additionally, it provides space for the elevator to enable longitudinal control.

3.1.1 Horizontal Tailplane Geometry

Due to the low cruise velocity and altitude of the UAV, as shown in Table 7 in Appendix A, it will operate at a Mach number of 0.1. Consequently, compressibility effects and the aerofoil's critical Mach number are not design constraints. Therefore, no sweep is added, as it would only reduce the horizontal lift curve slope [7].

Typically, taper is introduced into the horizontal tailplane to improve the structural properties of the empennage by moving the mass closer to the fuselage [7]. However, due to the boom configuration and the dual vertical tailplanes supporting the horizontal tailplane at either end, this is unnecessary. As a result, a taper ratio of 1 was chosen.

The aspect ratio was left open for mass optimisation, as discussed in section 3.3. Additionally, since tip stall and an elliptic lift distribution are not constraints for the horizontal tailplane, no twist was added [7].

3.1.2 Horizontal Tailplane Aerofoil

Symmetric or nearly symmetric airfoils are preferred for the horizontal tailplane to ensure predictable lift in both directions [8]. Therefore, the NACA 0009 was chosen for its symmetric profile and lower drag compared to the NACA 0010 or NACA 0012 [9].

3.2 Vertical Tailplane Design

Similarly to the horizontal tailplane, section 3.1, the vertical tailplanes role is to ensure stability and trim but in the lateral direction.

3.2.1 Vertical Tailplane Geometry

Given the low cruise Mach number of about 0.1, section 3.1.1, the vertical tailplane doesn't need sweep for to counteract compressibility effects. Although sweep could have helped by pushing the horizontal tailplane further back to generate a greater moment arm, this was deemed unnecessary as the boom length can varied to achieve suitable stability and control performance, section 3.3.

Additionally, the connection between the vertical tailplane and the booms has not been fixed, section 3, allowing the horizontal tailplanes height to be adjusted to mitigate deep stall effects. Consequently, no taper or twist were considered, ensuring that the boom connection placement has minimal impact on the vertical tailplane's performance. Once the required vertical height of the vertical taiplane has been found, discussed in section 3.3, it's vertical positon was changed such that the vertical distance between the horizontal tailplane and the wing is within the valid range which is described in [3] and depicted in Figure 12 in Appendix B.

Moreover, the chord of the vertical tailplane was selected to match that of the horizontal tailplane, aiming to simplify the design and prevent drag caused by mismatched chords intersecting.

3.2.2 Vertical Tailplane Aerofoil

Inverted U-tail empennage designs typically offer lower lateral stability due to the split vertical tailplanes reduced height [10]. Thus, a symmetric aerofoil with a higher lift curve slope than the NACA 0009 was first considered. However, the NACA 0009 was ultimately chosen because other aircraft components, such as winglets [11] and high-mounted wings, will aid the UAVs lateral stability and priority was given to the low drag characteristics of the NACA 0009.

3.3 Empennage Sizing Methedology

To size the chosen inverted U-tail empennage design, a methodology was developed to satisfy both longitudinal, section 3.3.1, and directional, section 3.3.2, stability while minimising total mass. This adaptable process allowed for adjustments to the UAVs mass distribution [12] and constraints throughout the design process whilst always ensuring that the minimum mass empennage design was found.

An overview of the design process is depicted in Figure 4, which shows how the wing x-position, w_x, and tailplane x-position, t_x, were considered free variables that were allowed to vary. For each wing and tailplane position, the required surface area of the horizontal and vertical tailplanes was determined to ensure stability and controllability. The mass of each configuration was then calculated and compared to previous iterations to identify the minimum mass configuration.

Various methods exist for estimating empennage mass, often relying on empirical data. However, these methods can be biased towards specific types of aircraft in their datasets. For instance, datasets like [13] with larger GTOWs tend to predict unrealistically low masses when applied to the REACH UAV. Similarly, datasets involving different empennage configurations [14] may not provide accurate estimates. Given the fixed design of the empennage, it was decided to use the same density, ρ_w, for the empennage as for the wing. Since the wing design was established early in the design phase with a conventional planform shape [15], using trend data from [13] was deemed appropriate to estimate the mass. This approach ensures that the empennage mass scales appropriately with the total tailplane surface area.

Equation (1) defines the objective function minimised to determine optimal tailplane sizing. Here, S_h and S_v represent the planform areas of the horizontal and single vertical tailplanes, respectively. Additionally, B_l denotes the required boom length for tailplane placement, and B_A denotes the cross-sectional boom area.

$$M_e = \min_{w_x, t_x} \rho_w \Big(S_h(w_x, t_x) + 2S_v(w_x, t_x) \Big) + \rho_B B_A B_l(t_x) \tag{1}$$

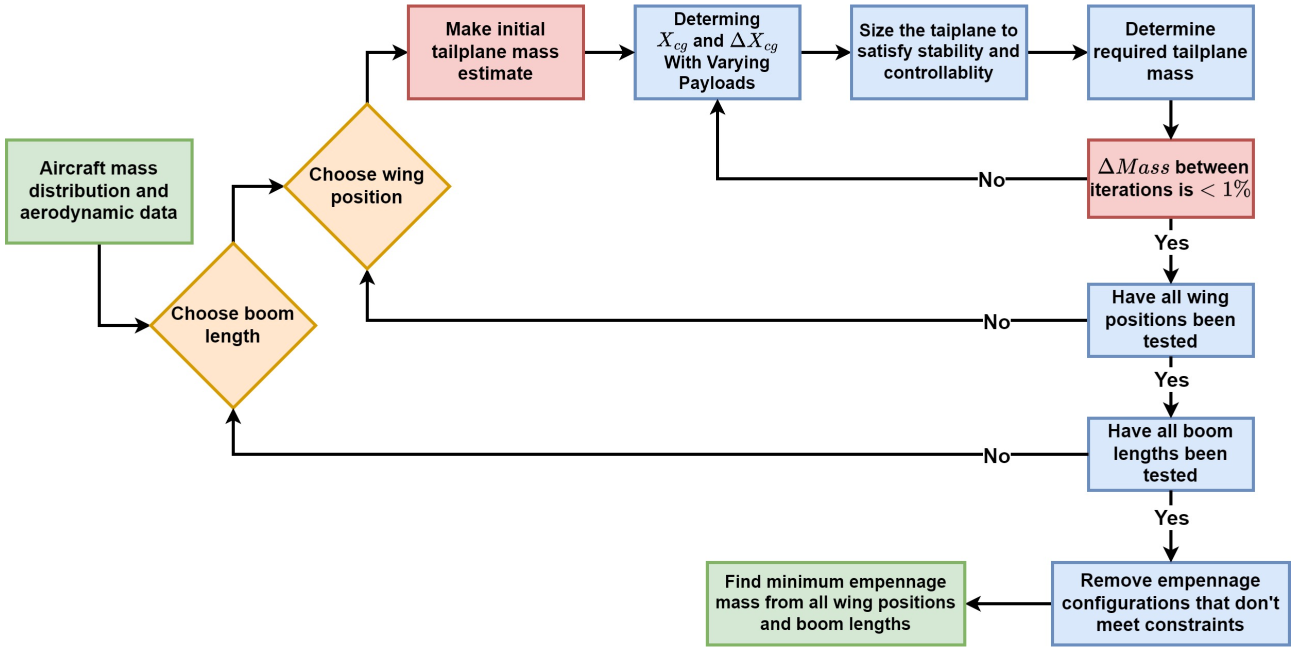


Figure 4. Flowchart diagram illustrating the methodology employed to size the horizontal and vertical tailplanes, and to position the wing and empennage along the x-axis.

3.3.1 Empennage Longitudinal Stability and Controllability

To properly size the horizontal tailplane, it must generate sufficient lift to ensure both longitudinal stability and control. For longitudinal stability, this means the UAV can return to its original position after a pitch disturbance. For controllability, it refers to how easily the UAV can be trimmed at various pitch angles.

Longitudinal static stability is determined through the aircraft's C_{m_α} which is the coefficient of pitching moment with respect to angle of attack. To achieve longitudinal stability, C_{m_α} must produce a restoring moment when the UAV undergoes a perturbation in pitch angle i.e. it must be negative. C_{m_α} can be calculated with Equation 2.

$$C_{m_\alpha} = \frac{\partial C_m}{\partial \alpha} = C_{m_{a,w}} + C_{m_{a,f}} + C_{m_{a,B}} + C_{m_{a,h}} < 0 \quad (2)$$

The derivatives $C_{m_{\alpha,w,f,B,h}}$ refer to the contributions from the wing, fuselage, booms and horizontal stabiliser respectively. Then by modifying the method described in [16] to include the contributions from the fuselage and booms, the minimum area ratio between the wing and horizontal stabiliser required to achieve stability can be determined using Equation 3.

$$\frac{S_h}{S_w} = \frac{C_{L_{\alpha,w}}(\overline{x_{cg} - x_{w,ac}}) + C_{m_{\alpha,f}} + C_{m_{\alpha,B}}}{C_{L_{\alpha,h}} \cdot \eta_h \cdot (1 - \frac{\partial \epsilon}{\partial \alpha}) \cdot (\frac{l_h}{\bar{c}} - \overline{x_{cg} - x_{w,ac}})} \quad (3)$$

Where $C_{L_{a,w}}$ and $C_{L_{a,h}}$ are the lift curve slopes of the wing and horizontal stabiliser, $\overline{x_{cg} - x_{w,ac}}$ is the difference between the longitudinal centre of gravity position x_{cg} and the wing's aerodynamic centre $x_{w,ac}$ over the wing's mean aerodynamic chord $\bar{c}$, η_h is the dynamic pressure ratio, predicted to be 1 due to the inverted U-tail design and the use of a pusher propeller [4], l_h is the distance between the wing and horizontal stabilisers aerodynamic centres and $\frac{\partial \epsilon}{\partial \alpha}$ is the downwash derivative, calculated using Equation (18) in Appendix B. To determine the moment contribution from the fuselage $C_{m_{a,f}}$ and booms $C_{m_{a,B}}$, the Multhopp strip-integration method described in [17] is employed, as expressed in Equation 4.

$$C_{m_{\alpha,f,B}} = \frac{1}{S_w \bar{c}} \sum_{i=1}^{n} w^2(x_i) \frac{\partial \bar{\epsilon}}{\partial \alpha}_i \Delta x_i \tag{4}$$

This method calculates the impact of the booms and fuselage on C_{m_α} by treating them as objects extending forward or aft from the wing by a distance x. The width of the booms and fuselage is accounted for using the variable w, and $\frac{\partial \bar{\epsilon}}{\partial \alpha}$ represents the downwash derivative, which is calculated differently depending on whether the protrusion is in front or behind the wing. A diagram of this method and more detailed equations can be found in Figure 14 in Appendix B.

Similarly, the method from [16] is adapted to incorporate the fuselage and booms when calculating the minimum horizontal stabiliser area ratio for controllability, Equation 5. Ensuring, $C_m = 0$ at trim can be achieved.

$$\frac{S_h}{S_w} = \frac{C_{L,w}}{C_{L,h} \cdot \eta_h \cdot \frac{l_h}{\bar{c}}} (\overline{x_{cg} - x_{w,ac}}) + \frac{C_{m,w} + C_{m,T} + C_{m,f} + C_{m,B}}{C_{L,h} \cdot \eta_h \cdot \frac{l_h}{\bar{c}}} \tag{5}$$

The variables $C_{m,w}$, $C_{m,T}$, $C_{m,f}$, and $C_{m,B}$ denote the moments generated by the wing, engine thrust, fuselage, and booms during steady level flight at cruise. It was assumed that the angle of attack during this phase would be approximately 1 degree to minimise drag [7]. Additionally, it was assumed that the wing's coefficient of lift, $C_{L,w}$, matches the C_L required for cruise, and the horizontal tailplane's coefficient of lift was taken as -0.5, as suggested by [16]. This assumption was considered reasonable because the horizontal tailplane is expected to provide negative lift for trim purposes and this value is within the linear range of the lift curve slope for the chosen NACA 0009 aerofoil [9].

3.3.2 Empennage Directional Stability

Directional stability, like longitudinal stability, ensures the UAV can return to its original flight path after a disturbance. It concerns the aircraft's ability to recover from a sideslip induced by a disturbance in the yaw axis. For static stability, the yaw moment coefficient with respect to sideslip, C_{n_β}, needs to be greater than zero. From [17], a typical value for C_{n_β} is approximately 0.057 rad^{-1}. Therefore, this value is employed in subsequent equations to size the vertical stabilisers.

Typically, the most limiting scenario for lateral stability would be the one-engine-inoperative case. However, this was disregarded because the UAV has only one engine.

A technique from [16] is used to calculate the minimum vertical tailplane area ratio required to achieve directional stability, which is expressed by Equation 6.

$$\frac{S_v}{S_w} = \frac{C_{n_\beta} - C_{n_{\beta,f}}}{-C_{Y_{\beta,v}}} \frac{b}{l_h} \tag{6}$$

Noting that $C_{Y_{\beta,v}}$ is just the lift curve slope of the vertical stabiliser but in the lateral direction and $C_{n_{\beta,f}}$ denotes the fuselages contribution to the yawing moment derivative with respect to sideslip and can be calculated with Equation (22) in Appendix B.

Note, the UAV has two vertical stabilisers thus the value obtained for S_v is halved to get the platform area for each individual vertical stabiliser.

3.3.3 Empennage Sizing Constraints

Once all wing and tailplane positions were sized, some had to be discarded due to conflicts with exterior constraints.

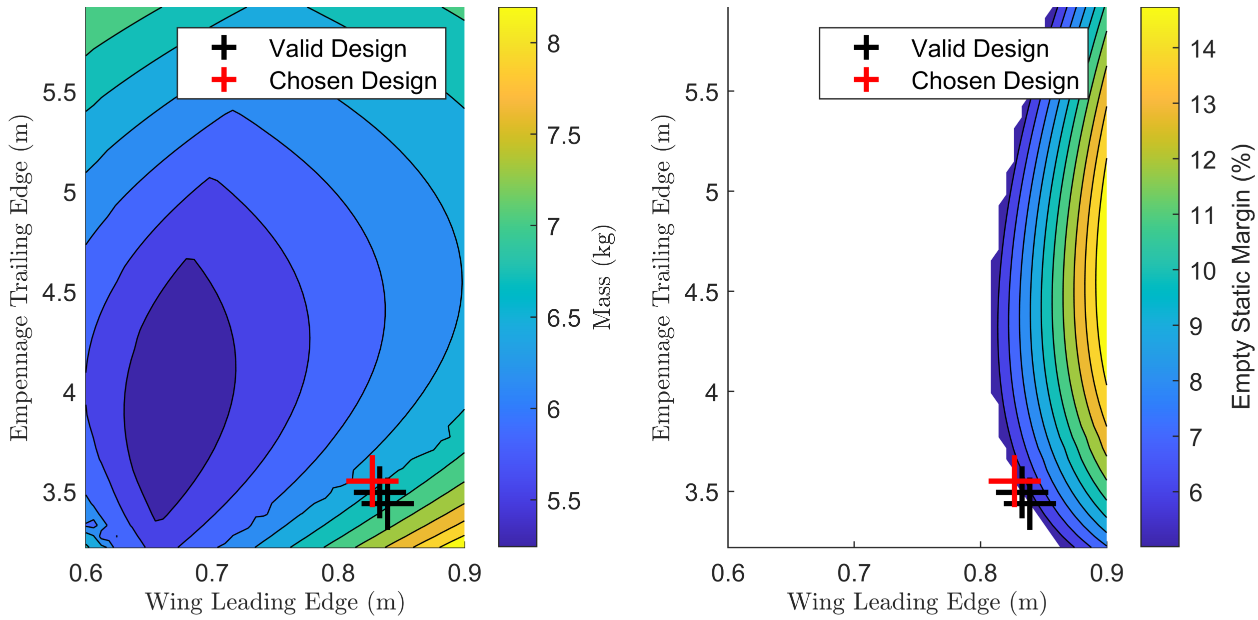
First, the wing leading edge was restricted to a position between 0.6m and 0.9m from the front of the fuselage. This aimed to push the wing towards the rear of the fuselage, eliminating the need for additional structural supports for the empennage. Details on this can be found in [18]. The position was also chosen to ensure that the wake and intake from the fore and aft vertical rotors were not obstructed by the wing, and that there was at least a 0.1m gap between any wing edge and the rotor tips to minimise interference effects.

Further more, the vertical height to the horizontal tailplane was enforced to be greater than the maximum height of the pusher propeller tip. This was to try and reduce the impact the propeller wake has on the elevator control surface [4]. Similar precautions were not taken for the vertical stabilisers because each was positioned at least 0.5m away from the propeller tips, due to the boom placement.

On top of this, it was deemed appropriate to ensured that the aircraft would have a static margin greater than 5% of the mean aerodynamic for all flight mass distribution, ensuring longitudinal stability and controllability as discussed in section 3.3.1 [16]. Therefore, particular attention was given to the extreme case of flying without fuel and payload, where the center of gravity would be furthest aft and closest to the UAV's neutral point [12].

The final constrains was set for the tailplane position. To allow for adequate placement of the landing gear such that a landing with 10° of tilt could be achieved [19] and to limit the structural strain on the booms.

3.4 Empennage Sizing Results



(a) Empennage total mass.

(b) Zero fuel and no payload static margin.

Figure 5. Results from the empennage sizing optimisation. Figure 5a illustrates the variation in mass of the empennage depending on the position of the wing and empennage. Figure 5b showcases the valid wing and empennage positions that produce static margins of greater than 5% of the mean aerodynamic chord when the UAV has no fuel or payload.

The results from combining together the sizing methodology, stability criteria and constraints is presented in Figure 5. These plots highlight how constraints significantly limit the number of viable empennage sizes. Out of 2500 different combinations of empennage boom lengths and wing positions, only three valid configurations were identified. Notably, the static margin constraint under empty flight conditions (zero fuel and payload) proved particularly restrictive, as depicted in Figure 5b. Suggesting that in order to have a more mass optimised empennage the x_{cg} of the empty fuel empty payload UAV should be moved further forward towards the front of the fuselage. Thus allowing the wing leading edge position to be shifted forward, achieving a reduced empennage mass Figure 5a. More results from the empennage sizing in regards to the span and vertical stabiliser span can be found in Figure 11b and Figure 11a in and Appendix B.

Table 2. Summary of finalised empennage parameters. Where T.E represents the trailing edge measured from the start of the fuselage, V_c the associated volume coefficients, defined Appendix B, and i the setting angle.

Surface	Area (m^2)	Chord (m)	Span (m)	AR	Mass (kg)	T.E. (m)	V_c	i°
Horizontal	1.01	0.44	2.30	5.23	4.00	3.55	0.69	-1.5
Vertical	0.22	0.44	0.49	1.11	0.85	3.55	0.04	0

Using the design point from the sizing methodology, the empennage dimensions could be finalised and are summarised in Table 2. From these values, the rudder and elevator control surfaces could also be sized. For the rudder the chord fraction of the vertical stabiliser is proposed to be between 0.15 and 0.4 and for the elevator the chord fraction between 0.2 and 0.4 from [20, 7]. Therefore, a chord ratio of 0.3 was chosen for both the rudder and elevator because of the matching tailplane chords.

The elevator deflection angle was set to the maximum feasible at $\pm 30^{\circ}$ to minimise the tail stall angle of attack [21], aiding controllability during transitions between fixed-wing and quadcopter modes, where high angles of attack are expected [22]. In contrast, rudder deflection was limited to $\pm 25^{\circ}$ [23]. Additionally, both control surfaces were positioned with a 0.1m gap before reaching the junction between the vertical and horizontal tailplane. This arrangement prevents interference between the elevator and rudder at maximum deflection and allows the empennage edges to be curved for drag reduction, as discussed further in [11]. The final tailplane dimensions are depicted in Figure 6

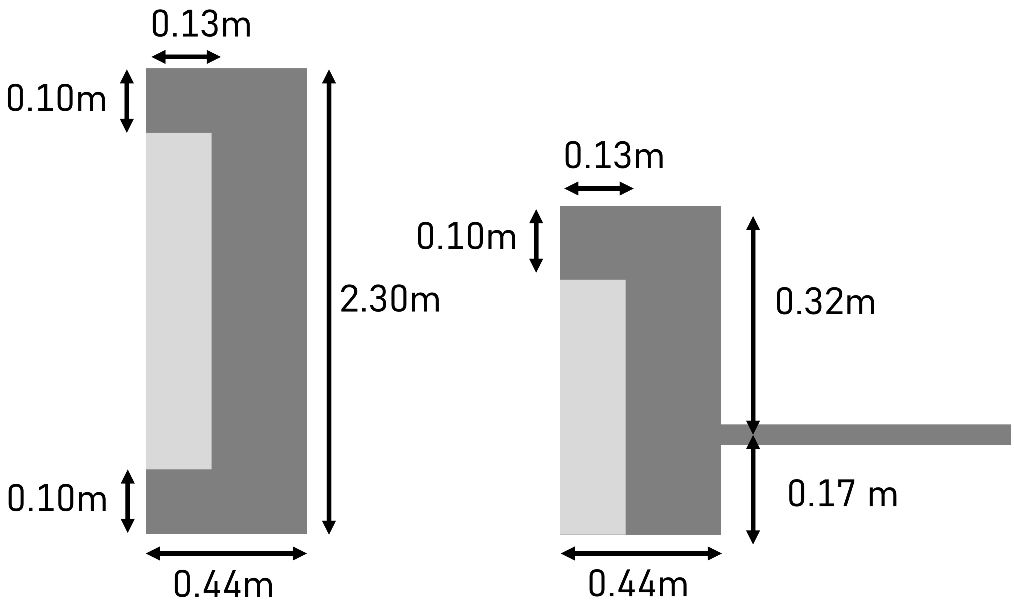


Figure 6. Final tailplane design with associated dimensions. The dark grey represents the fixed surfaces of the horizontal (left) and vertical (right) stabilisers where as the light grey represents the elevator and rudder control surfaces.

When comparing the empennage design results to those of other aircraft, the horizontal V_h and vertical V_v tailplane volume coefficients, Table 2, are found to be within the typical range of 0.5 to 1 for V_h and 0.02 to 0.05 for V_v. Suggesting that the methodology described in section 3.3 generates a feasible configuration.

4 Static Stability

To validate the static stability of the UAV, two methods were employed: a vortex lattice method (VLM) solver and analytical equations based on empirical data. The Athena Vortex Lattice (AVL) software was utilised for the VLM solver due to its strong performance in aerodynamic predictions [24]. Additionally, AVL offers good flexibility by allowing interfacing through configuration text files. Figure 15 in Appendix C displays the model of the UAV geometry in AVL. The equations from [17] were used to calculate the stability derivatives analytically. The results of both can be found in Table 3.

Table 3. Comparison of results obtained from AVL and analytical/empirical equations [17] in regards to the static stability of the UAV. Note, all the derivatives are measured in $rads^{-1}$.

Source	C_{L_α}	C_{Y_β}	C_{m_α}	C_{l_β}	C_{n_β}
Supply AVL	5.73	-.330	-.693	-.052	.035
Supply Analytical	5.44	-.262	-.620	-.085	.064
Supply Difference (%)	-5.06	-20.6	-10.5	63.5	82.9
Loiter AVL	5.75	-.331	-1.21	-.055	.037
Loiter Analytical	5.43	-.262	-1.07	-.087	.067
Loiter Difference (%)	-5.56	-20.8	-11.6	58.1	81.1
Empty AVL	5.75	-.330	-.284	-.053	.031
Empty Analytical	5.44	-.262	-.214	-.085	.062
Empty Difference (%)	-5.39	-20.6	-24.6	60.4	100

The results indicate the UAV is statically stable, meeting the criteria $C_{m_\alpha} < 0$, $C_{l_\beta} < 0$, and $C_{n_\beta} > 0$ in both AVL and analytical methods. AVL and analytical values align more closely for longitudinal derivatives (C_{L_α} and C_{m_α}) than for lateral derivatives (C_{Y_β}, C_{l_β}, and C_{n_β}). The greatest discrepancy is in the yawing moment with respect to sideslip, C_{n_β}, which was expected to be 0.057 rad^{-1} based on the analytical method used for sizing the vertical stabiliser, section 3.3.2. This is likely because the method used to size the vertical stabiliser is based on an analytical approach and therefore is slightly biased. Comparing from a different source, the lift curve slope C_{L_α} for the entire UAV, determined from CFD, was 5.72 rad^{-1} [25], indicating AVL's greater accuracy for this parameter. Overall, it was deemed appropriate to continue to use AVL to investigate the properties of the UAV and it will be the source of all UAV body axis derivatives.

Generally, the static stability derivatives remained consistent across different flight configurations, except for the pitch axis moment derivative with respect to angle of attack, C_{m_α}. This value changes between -1.21 rad^{-1}, -0.693 rad^{-1}, and -0.284 rad^{-1} for loiter, supply, and empty configurations, respectively. This significant shift is expected since C_{m_α} depends on the longitudinal x_{cg}, Equation (7). The positioning of the fuel tank and payload forward of the x_{cg} causes any added mass to decrease the x_{cg} and therefore decrease C_{m_α}. Table 8 in Appendix C showcases the positioning of the UAV components.

$$C_{m_\alpha} = -C_{L_\alpha} \cdot \frac{x_{np} - x_{cg}}{\bar{c}} \tag{7}$$

5 Modification to Lateral Body Axis Derivatives

Two main components not modeled in AVL and thus not considered in the stability derivatives are the booms and landing gear. Although the rotors and propeller are also not modeled, their impact is assumed negligible and just add parasitic drag. The following sections detail the effect of the skid landing gear and booms on the lateral body axis derivatives. Longitudinal derivatives, mainly affected by extra drag, are discussed in [6].

5.1 Effect of Booms

It was assumed that the booms would behave similarly to wing-mounted nacelles, as both are symmetrical along the x-axis and protrude from the wing. Thus, the UAV was considered to have two sets of nacelles: one set in front of the wings for the fore motors and one set behind the wings for the empennage and rear motors. Additionally, it was assumed that these nacelles would be cylindrical.

According to [26], the fuselage primarily affects the lateral derivatives related to sideslip velocity v. The roll rate p derivatives are dominated by the wing, and the yaw rate r derivatives by the wing and vertical stabilisers. Consequently, the booms were assumed to affect only the sideslip velocity lateral derivatives.

To measure the booms effect on the side force derivative due to sideslip velocity Y_v, the method from [27] was used, as shown in Equation (8). This method was chosen for its consideration of nacelle positions relative to the wing, unlike other methods [28], and is specific to propeller aircraft.

$$Y_{v_b} = -\frac{2}{3}\frac{K\pi B_h^2}{S_w} \qquad (8) \qquad\qquad n_{v_b} = Y_{v_b}\frac{B_{xm} - 0.46B_{fl}}{b} \qquad (9)$$

Where K is the fineness ratio of the nacelles and was altered depending on if the reference booms were fore or aft of the wing. The maximum cross sectional height of the booms is denoted as B_h.

To measure the booms contribution to the yawing moment derivative N_v, Equation (9) was used, again from [27]. The x-distance between the end of the booms and the c.g., B_{xm}, is positive for booms fore of the wing and negative for those aft. B_{fl} represents the x-distance from either the leading edge (fore booms) or trailing edge (aft booms) of the wing to the end of the booms.

The effect that the booms have on the rolling moment due to sideslip velocity L_v was accounted for using Equation (10), from [27].

$$l_{v_b} = -\frac{B_{zm}}{b}Y_{v_b} + 0.86(L_{v_b,T}) \qquad (10)$$

Where B_{zm} is the z-distance between the boom centre line and the c.g. which is positive if the booms are below the c.g. and negative if above. The term, $L_{v_b,T}$ is a theoretical values that takes into account the anti symmetric incidence loading and is dependent on AR, B_h, B_y and B_{zw}, [27].

As mentioned in [27], the results were used to gauge the magnitude of the booms impact on lateral derivatives. Table 4 shows that, due to the booms small cross-sectional height, high fineness ratio, and the wing planform area, their effect on lateral body axis derivatives is negligible. Thus, their contribution can be reasonably disregarded.

Table 4. Impact of booms on the lateral body axis coefficient derivatives with respect to sideslip velocity v. Note that the derivatives have been non-denationalised.

Derivative	C_{Y_v}	C_{n_v}	C_{l_v}
UAV (no booms)	-.330	.033	-.056
Just Booms	-2 $\times 10^{-7}$	3 $\times 10^{-8}$	-1 $\times 10^{-8}$
Impact (%)	≈ 0	≈ 0	≈ 0

5.2 Effect of Landing Gear

As discussed in [19], the REACH UAV will have a skid landing gear configurations. Since the landing gear will be attached below the fuselage it was assumed to be an extension of the fuselage with regards to the stability derivatives. Using this assumption, the landing gear would have the greatest effect on the yawing moment and side force moment derivatives with respect to side slip [26].

In regards to the yawing moment derivative N_v, the original value obtained from AVL was used in tandem with Equation (11) described in [26]. To find the non-dimensional coefficient n_f.

$$n_f = \frac{S_w b}{S_f l_f} C_{n_v} \quad (11) \qquad n_{v_l} = \frac{1}{2}\rho V^2 S_l l_l n_f \quad (12)$$

Where S_f is the side profile area of the fuselage and l_f is the length. Then by using the cross sectional properties of the landing gear instead of the fuselage, the landing gears contribution to the yawing moment derivative can be calculated via Equation (12). Here S_l and l_l now refer to the side profile area and length of the landing gear. A similar approach can be taken for the side force derivative y_f with Equation (13) and (14).

$$y_f = \frac{S_w}{S_f} Y_v \quad (13) \qquad Y_{v_l} = \frac{1}{2}\rho V^2 S_l y_f \quad (14)$$

Table 5 shows the landing gear has minor effect on lateral derivatives, especially compared to the differences between AVL and analytical/empirical values in Table 3. The assumption of shared coefficients between the landing gear and fuselage may be incorrect, potentially exaggerating the landing gear's impact.

Table 5. Impact of skid landing gear on the lateral body axis coefficient derivatives with respect to sideslip velocity v.

Derivative	C_{Y_v}	C_{n_v}
UAV (no skids)	-.330	.033
Just Skids	-.018	.001
Impact (%)	5.45	3.03

6 Lateral Dynamic Stability

Dynamic stability refers to an aircraft's ability to return to a steady state after a disturbance. For a UAV, analysing its lateral dynamic stability involves calculating the eigenvalues of its linearised motion equations. The three primary lateral modes: Dutch Roll, Roll Subsidence, and Spiral will be examined next, while the yaw independence mode is deemed neutrally stable due to its zero eigenvalue [29]. The physical interpretation of the yaw independence mode is that when the aircraft is perturbed in yaw the heading angle will move with it.

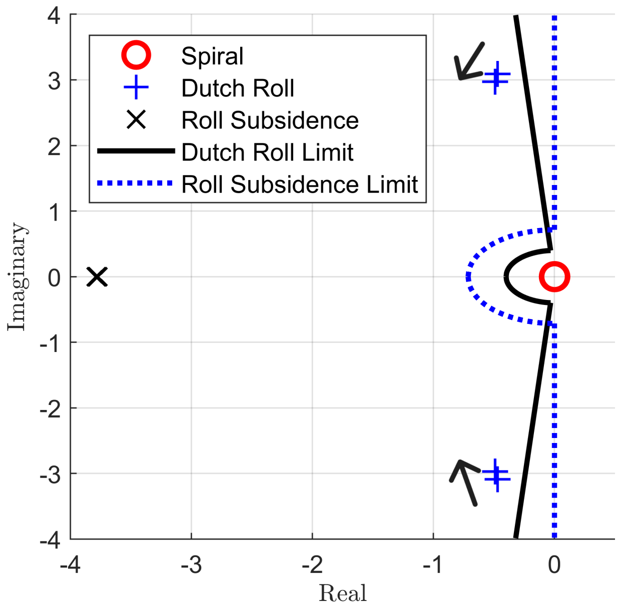
The UAV's dynamics were assessed for acceptability against the control criteria from AFFDL-TR-76-125 [30], which defines flying quality standards for remotely piloted vehicles. The criteria are summarised in an Table 6.

Table 6. Criteria set out by AFFDL-TR-76-125 [30] such that the lateral dynamic aircraft modes are clearly adequate for the mission flight phase. The natural frequency is denoted by w_n (rad/s) and the damping ratio by ζ.

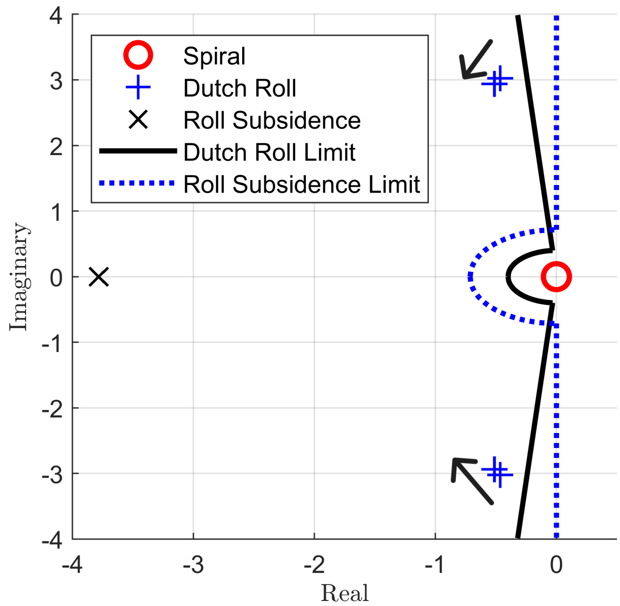
Mode	Roll Subsidence	Spiral	Dutch Roll
w_n	>.71	<.03	>.40
ζ	-	-	>.08
$w_n\zeta$	<0	-	<0

6.1 Varying Mass

During loiter and supply missions, the UAV's mass will significantly change, decreasing by 24.75% and 36.25%, respectively, due to fuel consumption and payload removal. These changes affect lateral stability minimally, as shown in Figure 7, with the dutch roll mode being most sensitive. Roll subsidence is excessively stable due to the dominant wing-induced rolling moment derivative L_p originating from the large wing, while minor effect in the spiral mode are detailed further in the Figure 16a and Figure 16b in Appendix D. All modes meet the requirements, Table 6 throughout.



(a) Loiter mission.



(b) Supply mission.

Figure 7. Root locus plots in Figures 7a and 7b show the lateral aircraft modes with varying masses due to fuel burn during loiter missions and payload unloading during supply missions. Note, arrow direction shows reducing mass.

6.2 Varying Cruise Velocity

The UAV's stability at cruise speeds above the 14.3m/s stall speed [25] is crucial for its transition between quadcopter and fixed-wing modes. Figure 8 shows lateral modes from stall to 60m/s, remembering that the target cruise speed is 41m/s. Both the Roll Subsidence and Dutch Roll linearly decrease in stability as the UAV's cruise speed decreases but are always within the suggested limits of AFFDL-TR-76-125 with the spiral mode seen in more detail in Figure 17 Appendix D also within the limits.

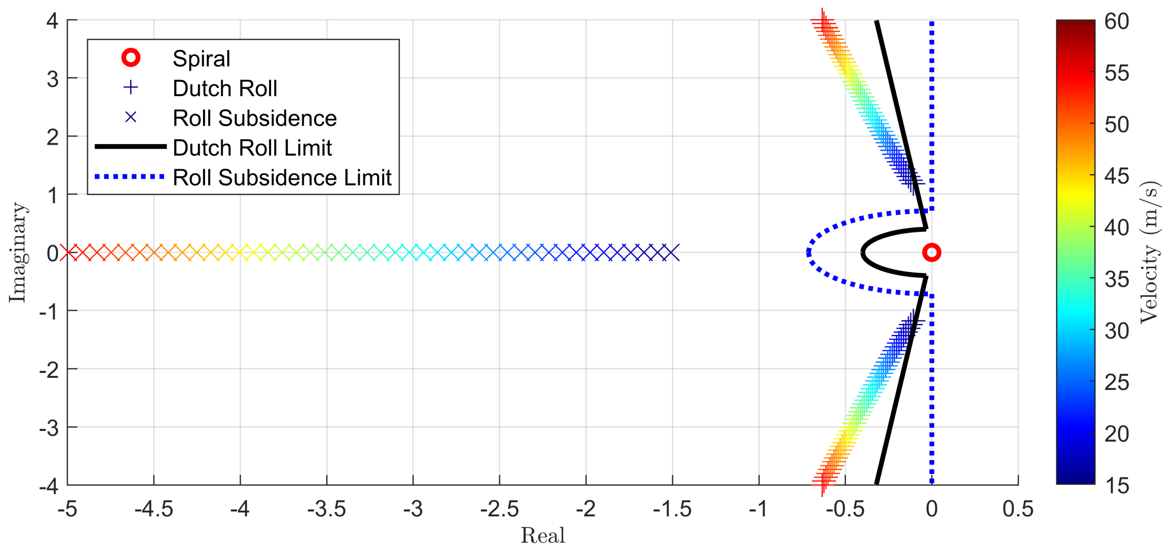


Figure 8. Root locus plot of the lateral aircraft modes at varying cruise speeds.

7 Discussion

7.1 Conclusion

In summary, the mass optimised inverted U-tail empennage design falls within the acceptable bounds for size and dimensions, as indicated by the tail volume coefficients V_h=0.69 and V_v=0.04 being between 0.5-1 and 0.02-0.05 respectively. On top of this, the static stability has been confirmed through analytical and computational methods to be stable. Noting that there is a substantial shift in the pitching moment coefficient C_{m_α}, ranging from -1.21 during loiter missions to -0.330 with no fuel or payload.

The investigation also revealed that the skid landing gear and booms barely affect the body axis lateral derivatives. This implies that in subsequent design versions, these elements might be deemed negligible and removed from consideration. This is particularly true since their influence arises from their physical dimensions and placement, rather than weight distribution, and remains constant regardless of mission type or variations in fuel and payload.

The Roll Subsidence, Dutch Roll, and Spiral modes of lateral dynamics have been confirmed to be stable or within the permissible ranges for a remotely operated unmanned aircraft according to FFDL-TR-76-125 standards. Adjusting the UAV's mass and cruise speed for analysis showed that these modes consistently stayed within acceptable limits at all speeds from stall to 60m/s and throughout fuel consumption during missions.

7.2 Next Steps

Although the methodology presented in section 3.3 effectively finds the minimum mass size and dimensions for the empennage, as illustrated in Figure 5a, it is extremely dependant on exterior constraints, especially when it comes to static margin constraints, depicted in Figure 5b, which are themselves sensitive to changes in x_{cg}. Therefore, when going through the conceptual design process, were the components haven't all been sized and finalised yet, the empennage size was consistently changing and it became a burden and a blocker to use this method, even if it was quick to find a solution. The mass distribution presented in Table 8 comes from a design freeze that was made such that the controls and structures teams had a base to work off. However, the final mass distribution for this first iteration can be found in [12]. Using this new mass distribution, the old empennage design is still valid, with the static margins now becoming 6.938%, 17.030% and 6.938% for the supply, endurance and empty missions respectively, as all of the static margins are above 5%, it is just no longer optimal. In order to improve the methedology, the design of the empennage should no longer be dependent on just the mass distribution as the input. Instead both the mass distribution and empennage sizing teams should choose a certain Δx_{cg} to work towards. Thus both the mass distribution and empennage sizing can be optimised and altered independently as long as the Δx_{cg} is maintained.

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Appendix

A Mission Profiles

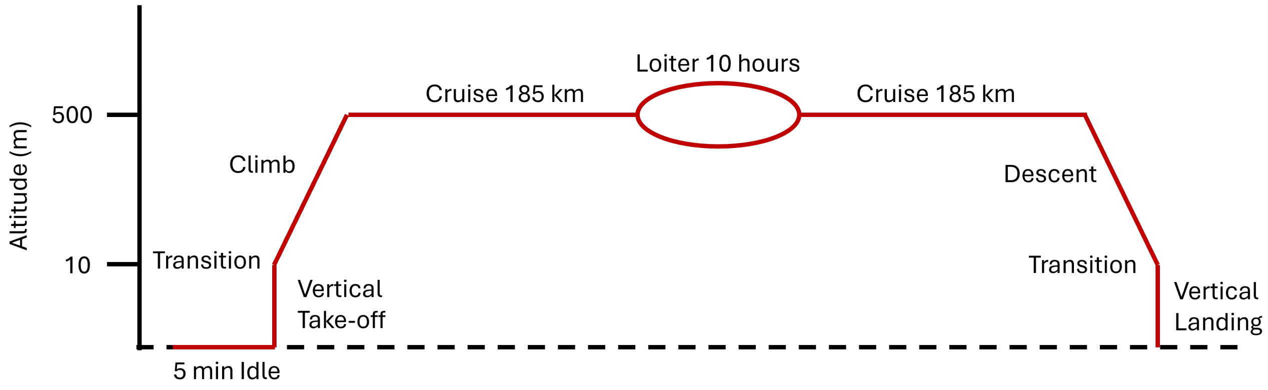


Figure 9. Endurance mission profile in which the UAV will carry a 20 kg payload.

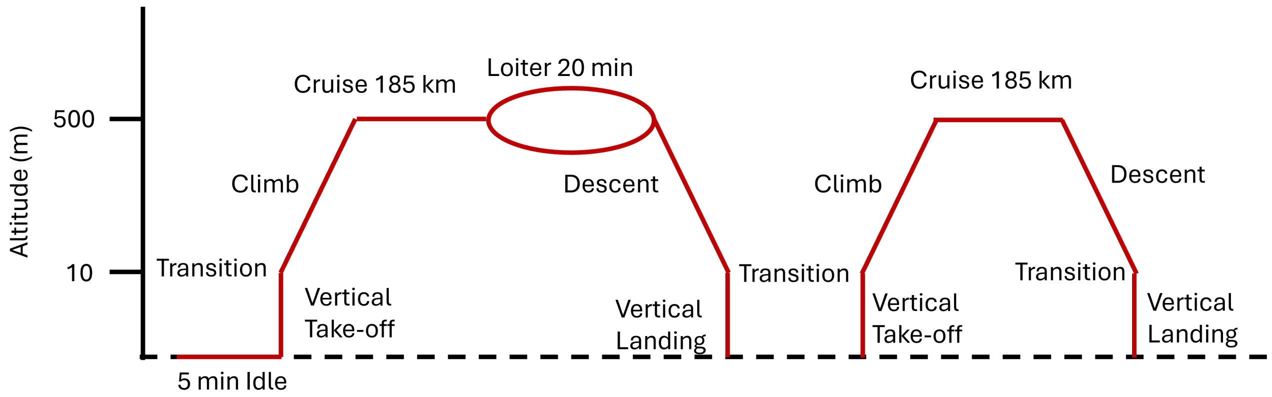
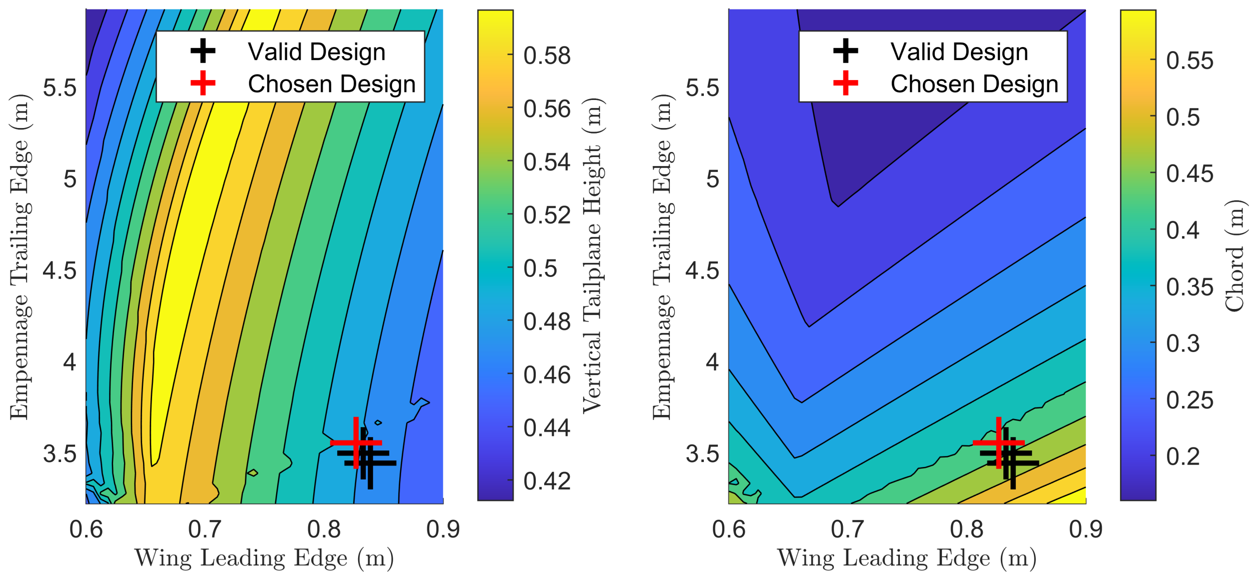


Figure 10. Transport mission profile where the UAV will carry a 50 kg payload for the initial 185 km, drop it off, and then fly an additional 185 km.

Table 7. Specifications and target parameters set out by the Vertical Flight Society [1].

Metric	Long Endurance	Supply Delivery
GTOW (kg)	160	160
Payload Mass (kg)	20	50
Loiter (hours)	10	0.33
Mission Radius (km)	185	185
Cruise Velocity (m/s)	41	41
Cruise Altitude (m)	500	500

B Empennage



(a) Empennage total mass. **(b)** Zero fuel and no payload static margin.

Figure 11. Results from the empennage sizing optimisation for chord and vertical stabiliser span.

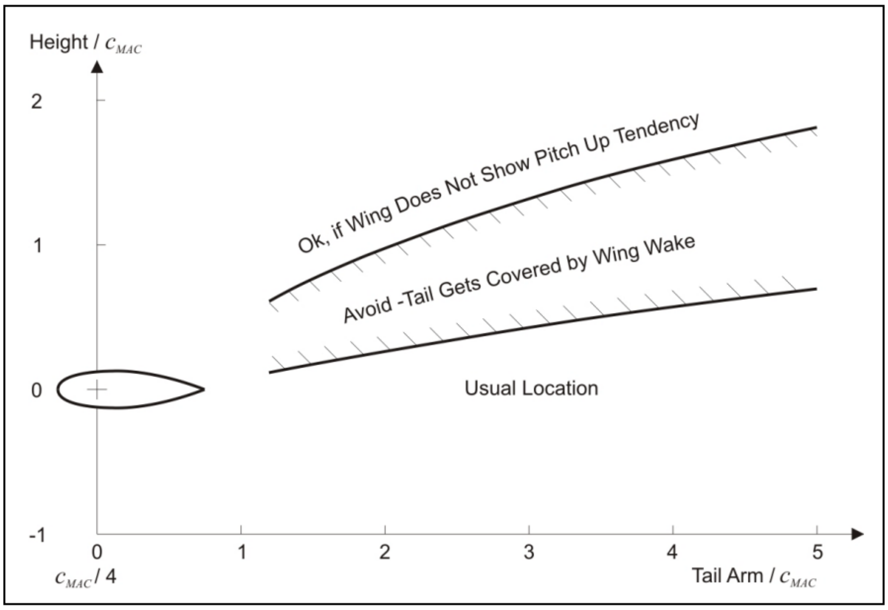


Figure 12. Horizontal tailplane positioning to avoid deep stall [3, 8].

Equations to calculate the downwash derivative with respect to angle of attack. All of the dimensions can be found with the aid of Figure 13.

$$K_A = \frac{1}{AR} - \frac{1}{1 + AR^{1.7}} \quad (15)$$

$$K_\lambda = (10 - 3\lambda)/7 \quad (16)$$

$$K_h = \frac{1 - |h_h/b|}{(2l_h/b)^{1/3}} \quad (17)$$

$$\frac{\partial \eta}{\partial \alpha} = 4.44(K_A K_\lambda K_h)^{1.19} \quad (18)$$

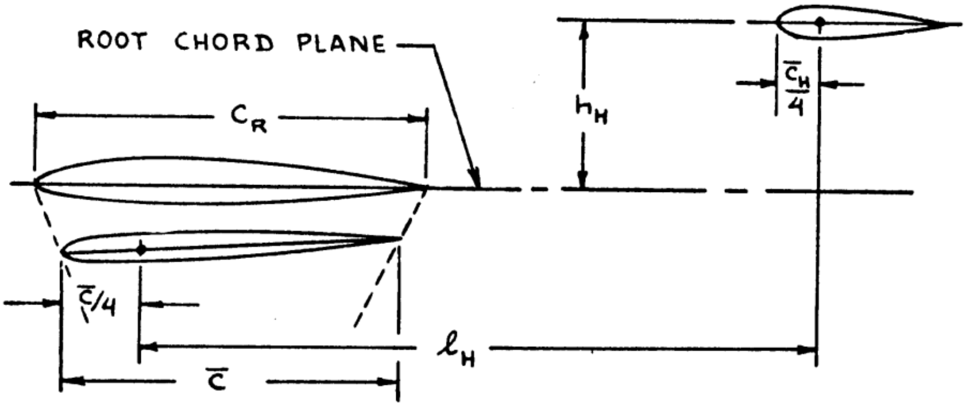


Figure 13. Dimensions associated with the wing and horizontal tail [17].

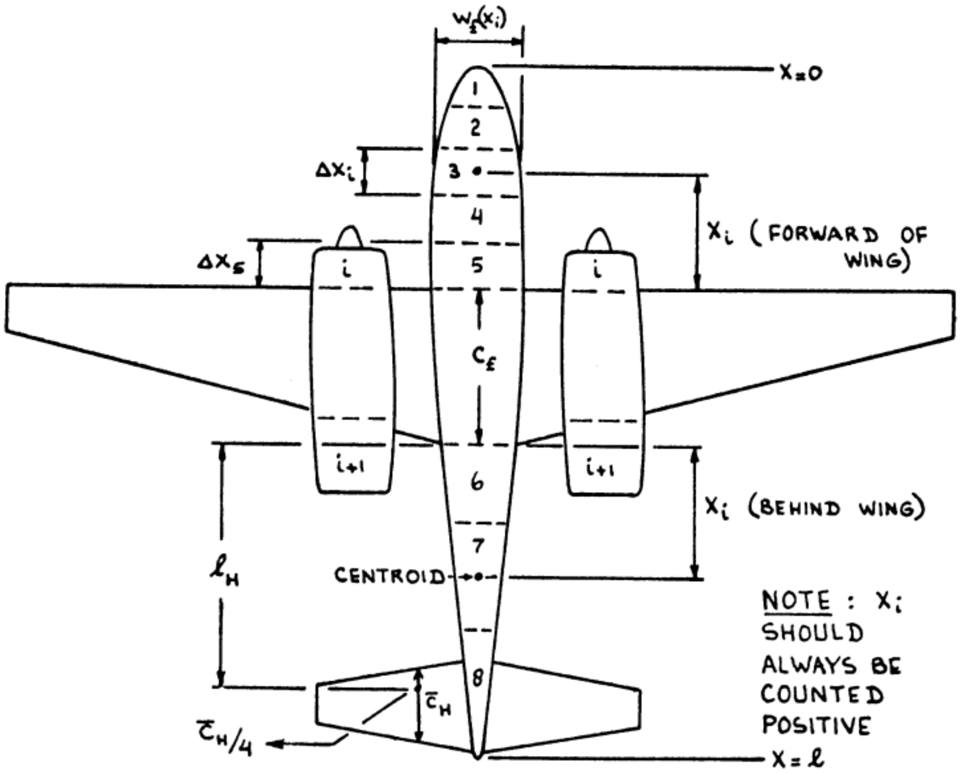


Figure 14. Dimensions associated with the fuselage and booms in reference to the Multhopp method of integration [17].

Equations to estimate the coefficient $C_{n_{\beta,f}}$ which denotes the yawing moment caused by a sideslip angle due to the aerodynamic qualities of the fuselage [16].

$$Re = \frac{Vl_f}{\upsilon} \tag{19}$$

$$K_{r,l} = 0.46ln(\frac{Re}{1 \times 10^6}) + 1 \tag{20}$$

$$K_N = 0.01\Big(0.27\frac{x_m}{l_f} - 0.168ln(\frac{l_f}{d_f}) + 0.416\Big) - 0.0005 \tag{21}$$

$$C_{n_{\beta,f}} = \frac{-360}{2\pi}K_N K_{r,l}\frac{l_f^2 d_f}{S_w b} \tag{22}$$

To non-dimensionalise the horizontal and vertical stabilsiers, Equations 23 and 24 can be used.

$$V_v = \frac{S_h l_h}{S_w \bar{c}} \tag{23}$$

$$V_h = \frac{S_v l_v}{S_w b} \tag{24}$$

C Static Stability

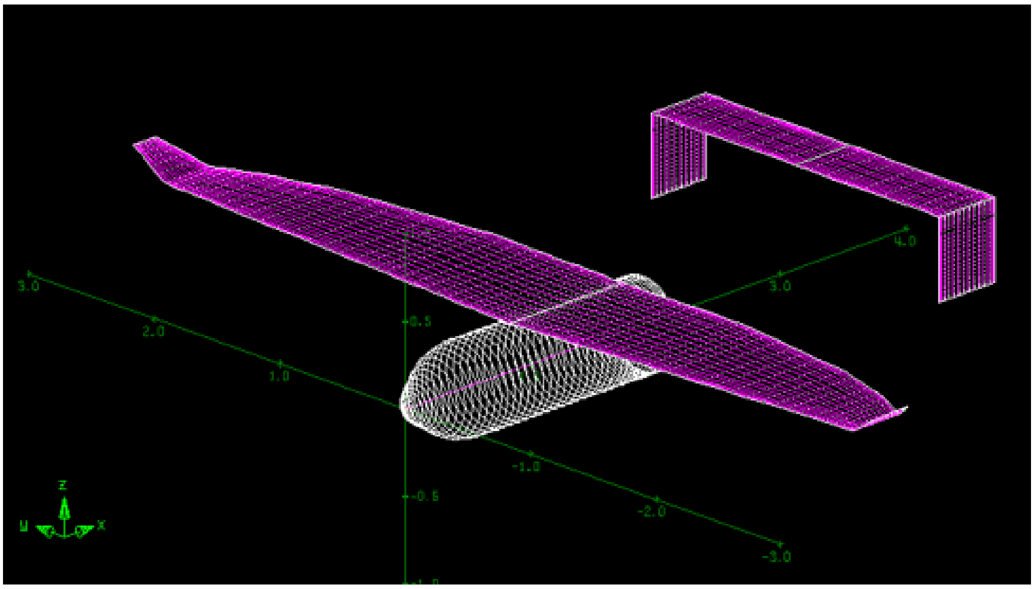
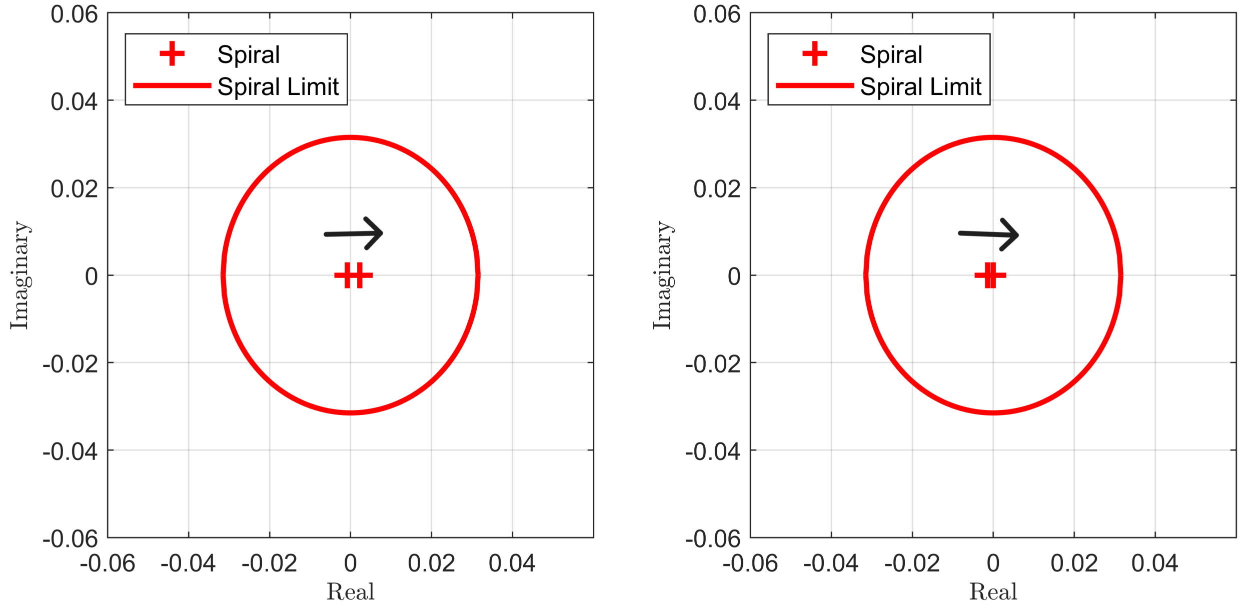


Figure 15. Geometry of the UAV in AVL.

Table 8. Mass distribution within the UAV.

Component	Mass (kg)	x_{cg} (m)	$y_{c}g$ (m)	z_{cg} (m)
wing	16.72	1.15	0	0.28
propeller	0.56	1.9	0	0
fuselage	15.5	0.95	0	0
landing gear right	1.75	1.1	0.25	-0.4
landing gear left	1.75	1.1	-0.25	-0.4
battery	5.4	0.31	0	0
e fuel	8.78	0.76	0	0
engine	11.5	1.79	0	0
gernerator	0.8	1.79	0	0
motor1	3.1	0.2	1.25	0.28
motor2	3.1	0.2	-1.25	0.28
motor3	3.1	2.32	1.25	0.28
motor4	3.1	2.32	-1.25	0.28
satellite	10.2	0.38	0	0
flight control	5	0.19	0	0
boom right	2	1.56	1.25	0.28
boom left	2	1.56	-1.25	0.28
empennage boom right	0.46	3.24	1.25	0.28
empennage boom left	0.46	3.24	-1.25	0.28
empennage	5.7	3.29	0	0.52
fuel system	2	1.56	0	0
payload	50	1.1	0	0
modular add on	7	1.1	0	0

D Lateral Dynamic Stability



(a) Loiter mission.

(b) Supply mission.

Figure 16. Root locus plots showcasing the effect of mass change on the Spiral mode. The arrows show the direction of reducing mass.

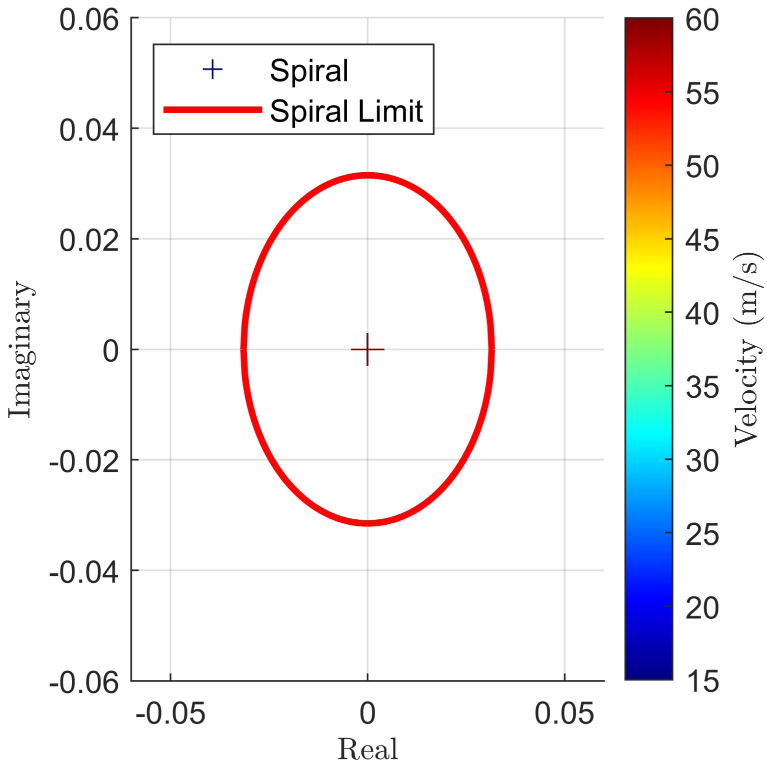


Figure 17. Effect of cruise speed on Spiral mode.